



THE COMPLETE GUIDE

Trace Chemo Waste Guide

Trace vs. bulk, the RCRA-empty rule, yellow containers, and USP 800 — how to handle chemo waste correctly.

DEA-REGISTERED

EPA-COMPLIANT

NO CONTRACT

CERTIFICATE OF DESTRUCTION

Compliance, made simple.

Trace chemo, contained & destroyed.

Chemotherapy waste splits into two very different streams: 'trace' chemo — RCRA-empty containers, PPE, and tubing with only residual drug — and 'bulk' chemo, which includes P-listed and other hazardous drug waste. Confuse them and you either over-incinerate or, far worse, put hazardous drug waste in the wrong container. This guide explains the trace-vs-bulk line, the RCRA-empty definition, the yellow-container system, and how USP 800 shapes safe handling.

The good news

Most day-to-day chemo waste is trace, and the yellow-container system makes it simple — as long as your team knows the one line between trace and bulk.

AT A GLANCE

Who regulates what.

Agency	What it covers in your practice
EPA	RCRA — the 'RCRA-empty' definition that separates trace from bulk, and P-listed bulk hazardous drug waste.
USP	USP General Chapter 800 — safe handling of hazardous drugs across receipt, compounding, administration, and disposal.
OSHA	Worker-protection expectations for hazardous drug exposure (PPE, containment).
DOT	Packaging and shipping papers for chemo waste in transit.

THE STREAMS

Every stream, in depth.

Below is each regulated stream your practice produces — what belongs in it, what does not, the rule behind it, how to dispose of it, and a tip to stay clean.

WASTE STREAM 01

Trace chemo waste

Materials that held or contacted a hazardous drug but are now RCRA-empty — the everyday chemo stream.

GOES IN THIS STREAM

- RCRA-empty vials, syringes & IV bags
- Tubing, gloves, gowns & contaminated PPE
- Empty containers meeting the RCRA-empty test

DOES NOT GO HERE

- P-listed drug waste or unused drug (bulk)
- More than de minimis residual drug
- Non-chemo sharps or RMW

THE RULE · EPA — a container is 'RCRA-empty' when all contents are removed by normal means and only de minimis residue remains; trace chemo is managed in the yellow stream.

DISPOSE WITH · Yellow trace-chemo containers, destroyed by permitted incineration.

Tip

An IV bag with visible drug still in it is not RCRA-empty — that's bulk, not trace.

WASTE STREAM 02

Bulk chemo & P-listed waste

Unused drug and P-listed chemo agents that exceed the trace threshold.

GOES IN THIS STREAM

- Unused/expired chemo drug
- P-listed agents (e.g. certain nitrogen mustards)
- Containers with more than de minimis residue

DOES NOT GO HERE

- RCRA-empty trace materials (yellow stream)
- Non-hazardous pharmaceutical waste

THE RULE · EPA RCRA — bulk and P-listed chemo is hazardous waste, managed and destroyed as such, not in the trace stream.

DISPOSE WITH · RCRA-hazardous (black-container) destruction by permitted incineration.

Tip

When a chemo item exceeds the trace threshold, it steps up to full hazardous-waste handling — don't leave it yellow.

WASTE STREAM 03

USP 800 handling

The standard that governs how hazardous drugs — including chemo — are handled through disposal.

GOES IN THIS STREAM

- Designated PPE for chemo handling
- Contained, labeled chemo waste at point of use
- A facility Assessment of Risk where applicable

DOES NOT GO HERE

- Chemo waste in ordinary trash or RMW
- Handling without the USP 800 controls

THE RULE · USP 800 — hazardous drugs are handled with defined PPE, containment, and disposal practices from receipt through waste.

DISPOSE WITH · Segregate at the point of use into the correct (trace vs. bulk) container, then destroy compliantly.

Tip

USP 800 is about worker and environmental safety end to end — disposal is the last link, not a separate silo.

YOUR CHECKLIST

Your trace chemo checklist.

- ☐ Staff trained on the trace-vs-bulk (RCRA-empty) line
- ☐ Yellow containers for trace chemo at the point of use
- ☐ Black/RCRA-hazardous containers for bulk and P-listed chemo
- ☐ USP 800 PPE and containment applied through disposal
- ☐ No chemo waste in ordinary trash or standard RMW
- ☐ Permitted incineration for destruction of both streams
- ☐ A Certificate of Destruction retained for every shipment

QUESTIONS ABOUT TRACE CHEMO WASTE

Trace chemo waste FAQ.

What's the difference between trace and bulk chemo waste?

Trace chemo is RCRA-empty materials — vials, syringes, IV bags, tubing, and PPE with only de minimis residual drug — managed in the yellow container stream. Bulk chemo is unused drug or P-listed waste that exceeds the trace threshold and is handled as RCRA-hazardous waste.

What does 'RCRA-empty' mean?

A container is RCRA-empty when all contents have been removed using the methods normally used for that container and only de minimis residue remains. That's the line that keeps most everyday chemo materials in the trace (yellow) stream.

What goes in the yellow container?

Trace chemo materials — RCRA-empty vials and syringes, empty IV bags, tubing, and contaminated PPE such as gloves and gowns. Unused drug and P-listed agents do not.

How does USP 800 relate to chemo waste?

USP General Chapter 800 sets safe-handling practices for hazardous drugs across their whole lifecycle — receipt, storage, compounding, administration, and disposal — including PPE, containment, and how waste is segregated.

How is chemo waste destroyed?

Both trace and bulk chemo waste are destroyed by permitted high-temperature incineration, with a Certificate of Destruction documenting the chain from your facility to destruction.

THE EASY WAY

Handle it with mail-back.

Give your team yellow containers for trace chemo and RCRA-hazardous containers for bulk, train them on the one line between the two, and let us destroy both by permitted incineration under USP 800-aligned handling — with a Certificate of Destruction every time.

QUICK REFERENCE · WHICH KIT FOR WHICH STREAM

Waste stream	The right path
RCRA-empty vials, PPE, tubing	Trace-chemo (yellow) container
Unused/P-listed chemo drug	RCRA-hazardous (black) container

Waste stream	The right path
Chemo-contaminated sharps	Trace-chemo path, not standard sharps
Non-chemo pharmaceutical waste	Pharmaceutical waste kit

Prefer pickup, or higher volume?

Oncology or infusion-scale chemo volume? We build USP 800-aligned programs with trace and bulk streams, PPE guidance, and scheduled service — ask for a quote.

WHY ONCOLOGY & INFUSION TEAMS CHOOSE EASY RX CYCLE

- No contract, no minimums — buy as you need
- Flat-rate pricing — no fuel or hidden surcharges
- One vendor for every stream you generate
- DEA-registered, EPA- & OSHA-aligned destruction
- A Certificate of Destruction on every single order

Ready to simplify your chemo waste?

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